

Hummingbird Midterm

Bug Investigation Report

Ticket #1 — "Server started on the wrong port"

Bug Location

File: `server.js` Line: 35

What's Broken

```
const port = process.env.APP_PORT;
```

Why It's a Bug

The port is read directly from the environment variable `APP_PORT` with no default fallback. If `APP_PORT` is not set, `port` is undefined. In Node.js, `app.listen(undefined)` does not crash — it silently binds to a random OS-assigned ephemeral port. The log outputs "listening on port undefined", and the server is unreachable on the expected port 9000.

Code Diff

Before:

```
const port = process.env.APP_PORT;
```

After:

```
const port = process.env.APP_PORT || 9000;
```

Ticket #2 — "Width is missing from metadata"

Bug Location

File: `clients/dynamodb.js` Line: 78-84

What's Broken

```
// createMedia saves width:
width: { N: String(width) },

// But getMedia never reads it back:
return {
```

```
mediaId,  
size: Number(Item.size.N),  
name: Item.name.S,  
mimetype: Item.mimetype.S,  
status: Item.status.S,  
};
```

Why It's a Bug

The width field is correctly stored in DynamoDB during upload via createMedia, but getMedia omits it from the return object. When a client calls GET /v1/media/:id, the response never includes width even though it exists in the database.

Code Diff

Before:

```
return {  
  mediaId,  
  size: Number(Item.size.N),  
  name: Item.name.S,  
  mimetype: Item.mimetype.S,  
  status: Item.status.S,  
};
```

After:

```
return {  
  mediaId,  
  size: Number(Item.size.N),  
  name: Item.name.S,  
  mimetype: Item.mimetype.S,  
  status: Item.status.S,  
  width: Number(Item.width.N),  
};
```

Ticket #3 — "Your redirect URL is broken"

Bug Location

File: `controllers/media.js` Line: `111`

What's Broken

```
res.set('Location', `${req.hostname}/v1/media/${mediaId}/status`);
```

Why It's a Bug

req.hostname in Express returns just the hostname (e.g., hummingbird-alb-xxx.elb.amazonaws.com) with no protocol prefix. The resulting Location header is not a valid absolute URL. Per RFC 7231, a Location header on a 3xx response should be a URI-reference with a scheme. Without http://, clients either fail to follow the redirect or interpret the value as a relative path.

Code Diff

Before:

```
res.set('Location', `${req.hostname}/v1/media/${mediaId}/status`);
```

After:

```
res.set('Location', `http://${req.get('host')}/v1/media/${mediaId}/status`);
```

***Note:** `req.get('host')` is used instead of `req.hostname` because it includes the port, which matters when the service is not running on port 80.*

Ticket #4 — "Download never redirects even when COMPLETE"

Bug Location

File: `controllers/media.js` Line: 108

What's Broken

```
if (media.status !== MEDIA_STATUS.PROCESSING) {
```

Why It's a Bug

This condition returns a 202 ("still processing") when the status is anything other than `PROCESSING`. That means: `PENDING` → 202 (correct by accident), `COMPLETE` → 202 (WRONG — should redirect), `ERROR` → 202 (also wrong). The only status that triggers the 302 redirect is `PROCESSING` — exactly backwards. The condition should check `!== COMPLETE` instead.

Code Diff

Before:

```
if (media.status !== MEDIA_STATUS.PROCESSING) {
```

After:

```
if (media.status !== MEDIA_STATUS.COMPLETE) {
```

Bonus — "Status never changes. No errors. Nothing."

Bug Location

File: `clients/dynamodb.js` Line: 154

Investigation Notes

Three DynamoDB functions operate on media records. `createMedia` and `getMedia` both use SK: 'METADATA' (uppercase), but `setMediaStatus` uses SK: 'metadata' (lowercase).

Function	PK	SK
<code>createMedia</code> (line 29)	<code>MEDIA#{mediaId}</code>	METADATA
<code>getMedia</code> (line 62)	<code>MEDIA#{mediaId}</code>	METADATA
<code>setMediaStatus</code> (line 154)	<code>MEDIA#{mediaId}</code>	metadata ← WRONG

DynamoDB keys are case-sensitive. When `setMediaStatus` runs an `UpdateItem` with SK: 'metadata', it targets a different item than the one created by `createMedia` (which used SK: 'METADATA'). Since there is no `ConditionExpression`, DynamoDB silently creates a new "phantom" item with the lowercase key. No error is thrown.

What Happens on PUT /resize:

- `setMediaStatus({ newStatus: 'PROCESSING' })` → creates NEW item (MEDIA#xxx, metadata)
- `copyMediaFile(...)` → succeeds
- `setMediaStatus({ newStatus: 'COMPLETE' })` → updates the NEW item to COMPLETE
- Response: { status: 'COMPLETE' } – looks correct
- GET /status calls `getMedia` → reads (MEDIA#xxx, METADATA) → original item → still PENDING

The original record is never touched. The status updates go to a phantom item.

Code Diff

Before:

```
SK: { S: 'metadata' },
```

After:

```
SK: { S: 'METADATA' },
```

Summary

Ticket	File	Line	Root Cause	Fix
#1	server.js	35	No default port fallback	<code>process.env.APP_PORT 9000</code>
#2	clients/dynamodb.js	78-84	width not returned in <code>getMedia</code>	Add width: <code>Number(Item.width.N)</code>
#3	controllers/media.js	111	Location header missing <code>http://</code>	Use <code>http://\${req.get('host')}/...</code>
#4	controllers/media.js	108	Wrong status comparison	Change to <code>!== COMPLETE</code>

Bonus	clients/dynamodb.js	154	SK casing mismatch	Change to 'METADATA'
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Verification

Ticket #1 — Server started successfully and logged "Example app listening on port 9000" (previously logged "port undefined").

Ticket #2 — getMedia now returns the width field in its response object, confirmed via GET /v1/media/:id.

Ticket #3 — The Location header in 202 responses now contains a valid absolute URL starting with http:// and including the host and port.

Ticket #4 — GET /v1/media/:id/download correctly returns a 302 redirect when status is COMPLETE, instead of always returning 202.

Bonus — setMediaStatus now targets the same DynamoDB item (SK: 'METADATA') as createMedia and getMedia, so status updates are applied to the correct record.

All endpoints were tested with curl and confirmed working as expected.